

Quiz — 1.4.1 Data Types

Name: _____ Date: _ **Mark:** ____ / 30

OCR H446 · Component 01 · 1.4.1 — show all working

Section A — Multiple choice (8 marks)

1. How many bits are represented by a single hexadecimal digit? A. 2 B. 4 C. 8 D. 16 Your answer: ☐
 2. What is the denary value of the unsigned binary number 00101101 ? A. 43 B. 44 C. 45 D. 47 Your answer: ☐
 3. In an 8-bit two's complement number, what is the place value of the most significant bit? A. +128 B. -128 C. +127 D. -127 Your answer: ☐
 4. Which operation has the effect of multiplying an unsigned binary number by 4? A. Logical shift right 2 places B. Logical shift left 2 places C. Arithmetic shift right 2 places D. Logical shift left 4 places Your answer: ☐
 5. In a floating-point number, which part determines the **precision**? A. The exponent B. The sign bit C. The mantissa D. The base Your answer: ☐
 6. A correctly normalised **positive** floating-point mantissa always begins with which bit pattern? A. 1.0... B. 0.0... C. 1.1... D. 0.1... Your answer: ☐
 7. The ASCII code for 'A' is 65. What is the ASCII code for 'D' ? A. 66 B. 67 C. 68 D. 69 Your answer: ☐
 8. Which statement about Unicode is correct? A. Unicode can only represent the English alphabet B. ASCII is a subset of Unicode, so Unicode is backward-compatible C. Unicode always uses exactly 7 bits per character D. Unicode cannot represent emoji Your answer: ☐
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Section B — Calculations & short answer (17 marks)

9. Convert the denary number 217 to 8-bit unsigned binary. [2]

Working space:

10. Convert the binary number 10110110 to hexadecimal. [2]

Working space:

11. Convert the hexadecimal number B7 to denary. [2]

Working space:

12. Using 8-bit two's complement, calculate 45 – 28 by adding the two's complement of 28. Show the binary working and state what happens to the final carry. [4]

Working space:

13. Normalise the following 8-bit signed floating-point number. The mantissa is `0.0110100` (one sign bit then the fraction) and the exponent is `0011` (4-bit two's complement). Give the normalised mantissa and the new exponent, and explain the rule used. **[4]**

Working space:

14. The 8-bit unsigned binary number `00010110` is shifted using a **logical left shift of 2 places**. Give the result and state the arithmetic effect. **[3]**

Working space:

Section C — Exam-style question (5 marks)

15. A programmer stores temperatures using floating-point representation with a fixed total number of bits.

(a) Explain the trade-off between allocating more bits to the mantissa versus more bits to the exponent. **[2]**

(b) The programmer says "I'll just use hexadecimal to store the numbers so they take less memory." Explain whether using hexadecimal reduces the amount of memory used, and give one genuine reason programmers use hexadecimal. **[3]**

Working space:

Answer key

Section A (8 marks — 1 each) 1. **B** — 1 hex digit = 1 nibble = 4 bits. 2. **C** — $32+8+4+1 = 45$. 3. **B** — MSB in two's complement is the negative place value -128 . 4. **B** — logical left shift n places = $\times 2^n$; left 2 = $\times 4$. 5. **C** — mantissa holds significant figures → precision (exponent sets range). 6. **D** — positive normalised mantissa starts `0.1...`. 7. **C** — 'A'=65, so 'D' = $65+3 = 68$. 8. **B** — first 128 Unicode code points match ASCII, so ASCII is a subset.

Section B

9. [2] `217` → binary. $217 = 128+64+16+8+1$. 128(1) 64(1) 32(0) 16(1) 8(1) 4(0) 2(0) 1(1) → check: $128+64+16+8+1 = 217$. ✓ Answer: **11011001**. Mark: correct method/subtraction (1); correct 8-bit answer (1).

10. [2] `10110110` → hex. Split into nibbles from the right: `1011 0110`. $1011 = 8+2+1 = 11 = \text{B}$. $0110 = 4+2 = 6 = \text{6}$. Answer: **B6**. Mark: correct nibble split/values (1); correct hex **B6** (1).

11. [2] `B7` → denary. $B = 11$, $7 = 7$. $(11 \times 16) + 7 = 176 + 7 = \text{183}$. Mark: $11 \times 16 = 176$ (1); $+7 = 183$ (1).

12. [4] $45 - 28 = 45 + (-28)$. $+45 = \text{00101101}$ ($32+8+4+1 = 45$). ✓ $+28 = \text{00011100}$ ($16+8+4 = 28$). ✓ -28 : invert `00011100` → `11100011`, then $+1 \rightarrow \text{11100100}$. Add:

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00101101   (45)
+ 11100100  (-28)
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1 00010001

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The carry out of the MSB is **discarded** (1 dropped), leaving 00010001. $00010001 = 16 + 1 = 17$. ✓ (45 - 28 = 17) *Mark: +45 and +28 in binary (1); correct two's complement of 28 = 11100100 (1); correct addition (1); discard final carry → answer 17 (1).*

13. [4] Mantissa 0.0110100, exponent 0011 (= +3). Normalisation rule: a positive number must have a mantissa beginning 0.1... (first significant bit immediately after the point). The current mantissa is 0.0110100 — it has one wasted leading 0 after the point. Shift the point **right by 1 place** to get 0.1101000. Moving the point right ⇒ the **exponent decreases** by 1: 0011 (+3) - 1 = 0010 (+2). Normalised mantissa = 0.1101000, exponent = 0010 (+2). (Value check: original = $0.011010 \times 2^3 = 0.40625 \times 8 = 3.25$; new = $0.110100 \times 2^2 = 0.8125 \times 4 = 3.25$. ✓ value unchanged.) *Mark: identifies mantissa must start 0.1 (1); shifts point right one place → 0.1101000 (1); exponent decreases to 0010 /+2 (1); states rule/value unchanged (1).*

14. [3] 00010110 logical left shift 2 places. Bits move left, 0s fill from the right, top bits lost: 00010110 → (1 left) 00101100 → (2 left) 01011000. Answer: 01011000. Effect: multiplied by $2^2 = \times 4$. Check: $00010110 = 22$; $01011000 = 64 + 16 + 8 = 88 = 22 \times 4$. ✓ *Mark: correct shifted bits 01011000 (1); 0s fill from right / bits lost stated (1); effect = $\times 4$ ($\times 2^2$) (1).*

Section C

15. [5] (a) **[2]** With a fixed total number of bits, giving more bits to the **mantissa** increases **precision** (more significant figures / smaller rounding error) but leaves fewer bits for the exponent, reducing **range** (1). Giving more bits to the **exponent** increases the **range** of values representable but reduces **precision** (1). It is a trade-off — you cannot increase both at once.

(b) **[3]** Storing the number "in hexadecimal" does **not** reduce memory (1): hex is only a human-readable shorthand for the underlying binary — the value is still stored as the same number of bits regardless of whether it is written in hex or binary (1). A genuine reason to use hex: it is a **compact, lossless representation of binary** that is easier for humans to read/write with fewer errors (e.g. memory addresses, colour codes, machine code) — 1 hex digit maps exactly to 4 bits (1).

Total: 8 + 17 + 5 = 30 marks.